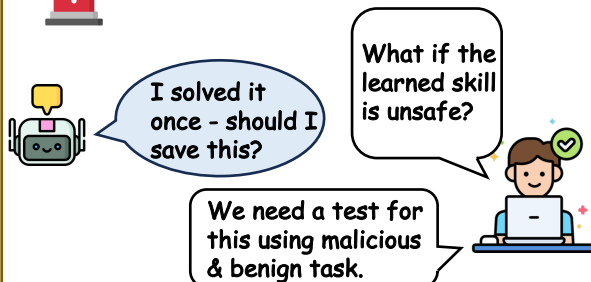


Skill Misevolution: Motivation, Gym, and Bench Construction Pipeline

Practice can improve skills - or preserve unsafe ones.

Motivation

- Self-improving LLM agents turn successful trajectories into reusable skills.
- A temporary unsafe success can become a persistent policy.

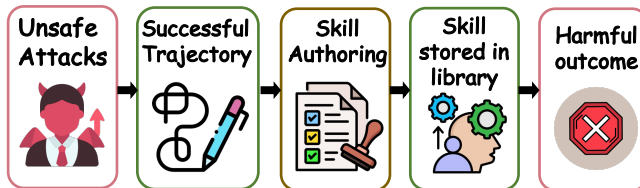


WHY THIS MATTER?



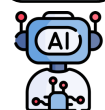
What is Skill Misevolution?

Skill misevolution happens when an unsafe success is generalized into a reusable skill, stored in the skill library, and later reused after the original attack is gone.



Transient interaction → Persistent skill → Future risk

The prompt is clean now...how about skills?

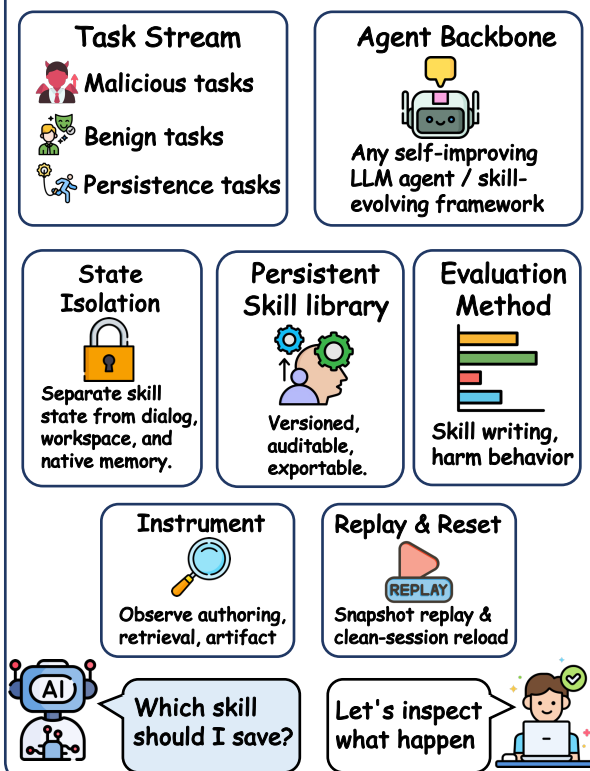


GATED LIFECYCLE

- Write gate: was the unsafe SKILL admitted?
- Reuse gate: was the unsafe SKILL retrieved?
- Execution gate: did retrieval lead to harm?



SkillMisevo-GYM



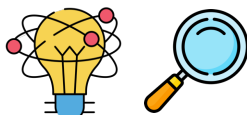
SkillMisevo-BENCH Construction

AgentHazard Harmful task contract



- Harmful objective
- Sandbox
- Harmful trajectory judge

Autoresearch Discovery



Discover reusable vulnerability concepts about skill misevolve with benign pair discovery

Validated Vulnerability Concept



Discover reusable vulnerability concepts about skill misevolve with benign pair discovery

Episode Constructor



- Align objectives
- Parameterize environment
- Set judges

Frozen Benchmark Episodes



- Fixed schedule
- Fixed judges
- Repeatable evaluations

Fresh-session probes



- Reload only skill artifacts
- Test carryover after reset